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Link to Github Repository:

https://github.com/jbecker20/DTSC2301/tree/5fe3c4279414486f47a1a1984b57e7e552322ea3/Project2

DTSC 2302

3/31/2026

Technical Report

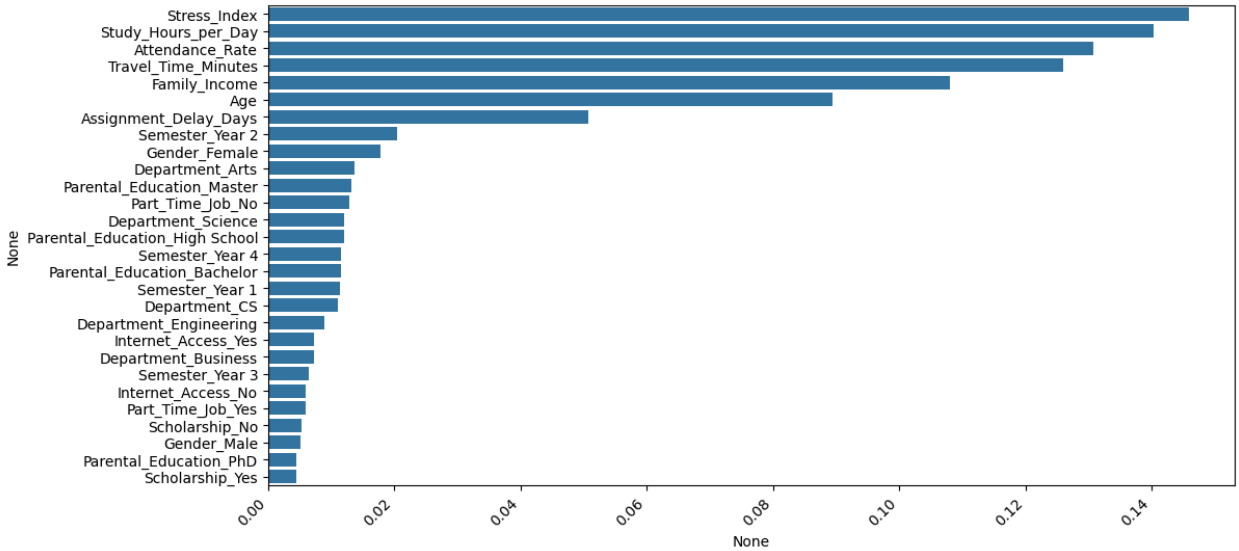
According to the Education Data Initiative 39% of students who pursue a bachelor's degree don't finish their degree. This made me want to dive into the question: what non-GPA factors cause a student to drop out of college? With the results from this question colleges can create aid to help students in ways other than just GPA to help increase retention in colleges and universities. This question can also help students visualize what factors are linked to dropping out of college.

After figuring out what question I was looking to explore I looked on kaggle to find a dataset that I thought would help me create a good model to predict these factors. I found a dataset labeled "Student Dropout Prediction Dataset" with different variables such as Family_Income, Age, Scholarship, Stress_Index, and Department. I knew this was a good dataset due to the variety of variables I would be able to use in my prediction models. First, I loaded my dataset into a jupyter notebook profile so I could explore the different values of my dataset. The first couple things I noticed just looking through the values is since this is a synthetically created dataset the data might not be 100% accurate and could also contain inaccurately processed information. I noticed there were a lot of missing values in multiple different columns. Another thing I saw was

the family income all seemed to have an extra 0 and it is supposed to be monthly so the lowest monthly family income being $25,000 doesn't make any sense. Something else that looked to be wrong with family income was there were over 3,000 students that all had $25,000 for their family income. Since this is a synthetic dataset I assumed that this data must be a mistake for both factors. To fix these issues with the income I divided everything by 10 so the values made a little more sense and then I created 2 datasets with one containing the 3000 students with family income of $25,000 and one that didn't have those students. To fix the missing values I looked into the distribution of the values and found that family income was skewed to the right meaning that the null values must be filled with the median. For the non categorical values besides family income I used the mean as they all had a normal distribution. For the categorical columns there is no central measure to fill the missing values so instead I would drop those rows that I wasn't able to fill. The final part of data processing I had to do before I could start creating models was use one hot encoding to make the categorical variables into numerical values so they can be used in a predictive model. This way the models can actually use these categorical variables to predict if a student would drop out.

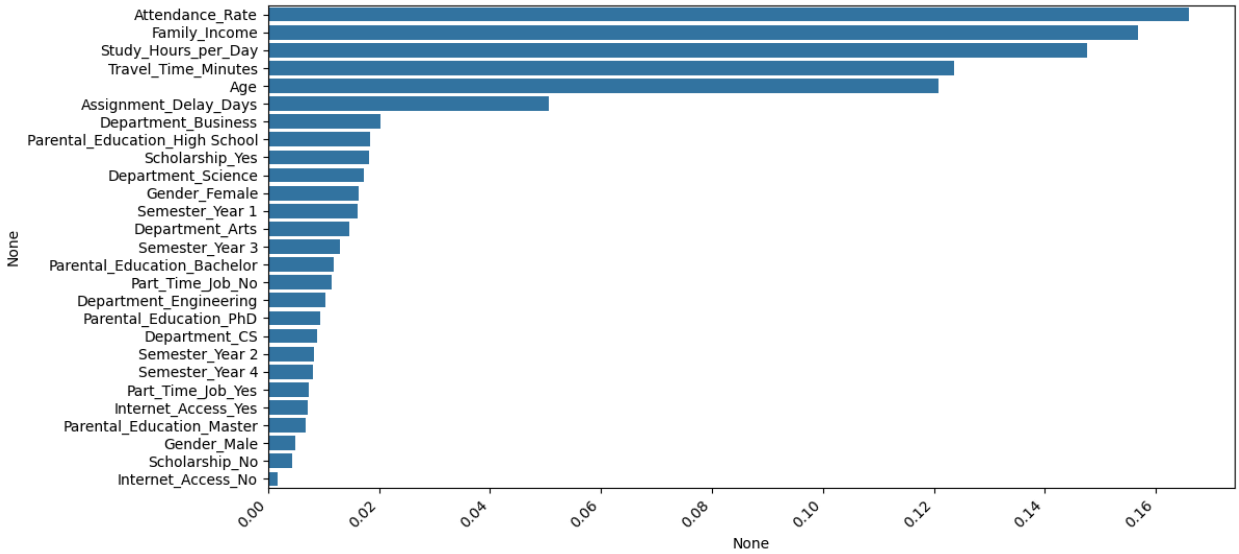
The first model I made I used the dataset that didn't contain the 3000 students that all had the same family income. With this model I was looking to see what the decision tree uses as the major factors to predict if a student will drop out or not. After setting my X to the dataset without dropout and y to the dropout variable I used the train test split function to make a training and testing set. I then fit my model using these training sets so that my model will be able to predict if a student will drop out or not.

After fitting the model I used the score function to test the model's accuracy and have a way to compare the accuracy of one model to another. The final thing I did with this first model was make a white box model and a bar chart to see what were the most important variables the model was using to predict the dropout variable.

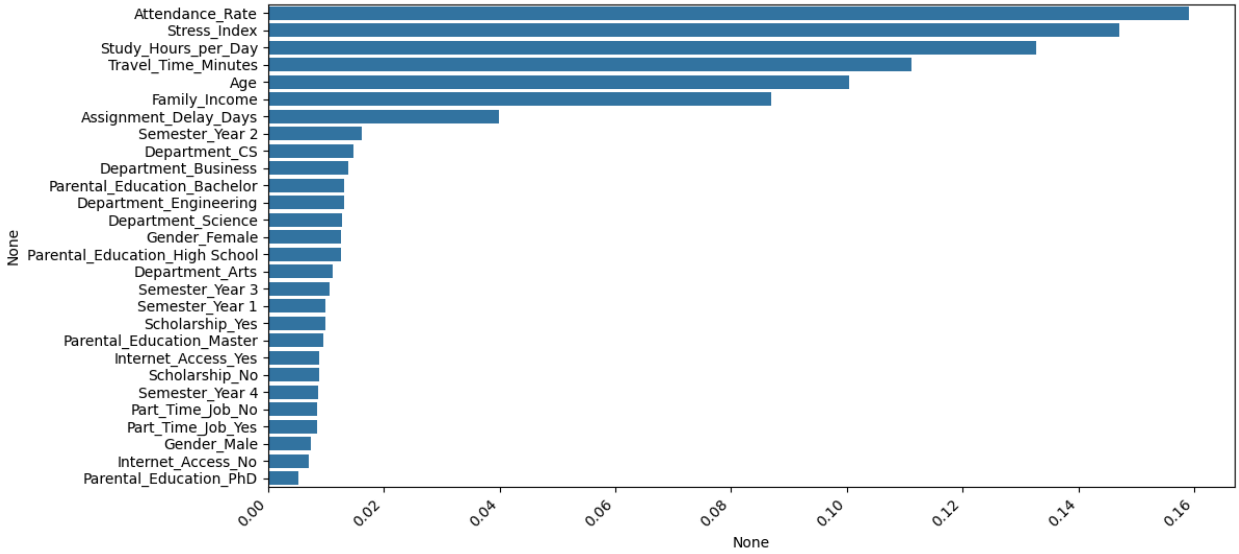


After seeing that stress index was the top variable that the model was using to predict dropout I decided to remove it and see how the model performed without stress index. Once again I used the dataset without the 3000 students and this time dropped the column containing stress index. Next I did another train test split with a 20% split to have a training and testing data set. Then I initiated a new decision tree classifier and fit it to the new training and testing dataset. After fitting the model I tested it using the score function to be able to compare it to model 1. I noticed that the accuracy score of model 2 was lower than model 1 which makes sense as I removed the main prediction variable from model 1. I created a bar plot to show what variables the 2nd model was using to predict the dropout variable and got a similar result to model 1 but this time

without the stress index.



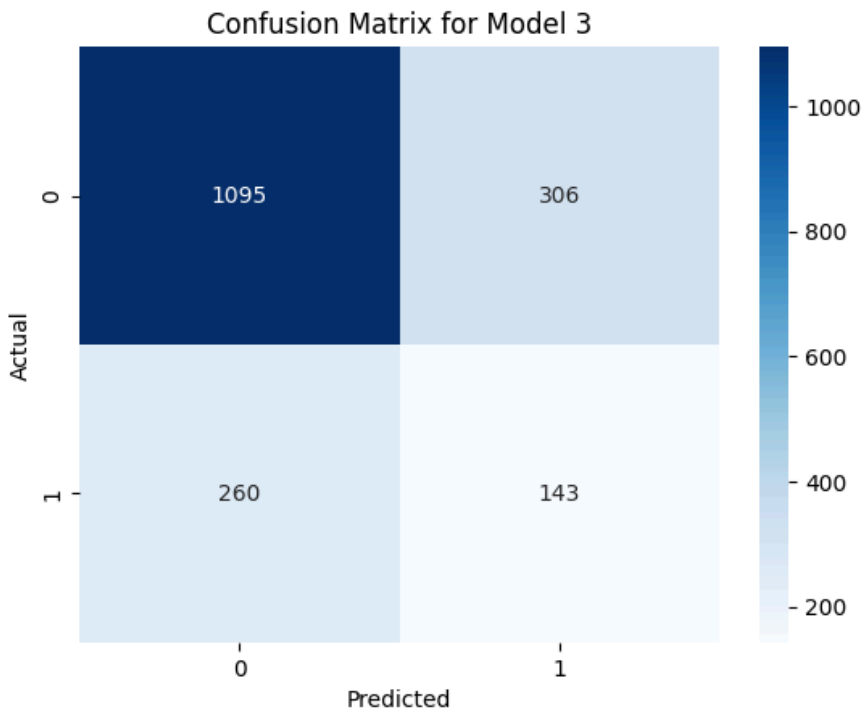
For the last model I brought the 3000 students I removed for the first 2 back into the dataset. Then I created a new X and y and created a training and testing dataset in similar fashion to the first 2 models. I then fit the last model to the new training and testing set and tested the score of the model afterwards. Once I finished the third model I created another bar plot and was surprised to see that attendance rate had become a larger factor than stress index had been for the other two models.



After I finished these 3 models I wanted to find a way to compare each of these models without just using the score function. I also wanted to see how these models compared for predicting false negatives as that is the most harmful to students. First I used the cross_val_score function to create 5 models of each and report what the average accuracy score was for each of the 3 models. Then I did the same for the recall score. Usually model 3 was the best when it came to average recall score which meant it was on average predicting the highest proportion of dropout cases compared to other models. For a final comparison I did the same test but this time for precision. Once again model 3 led in this category meaning of the dropouts it was predicting model 3 had the most true positives. This leads me to believe that removing the 3,000 students for having the same income was a bad decision as it likely had a larger effect on the results than I originally thought when removing those select students. I also learned from these tests that removing stress index from the also had a large effect on the models ability to predict dropouts. I also tested different depths to see if that had any effect on the models and after using the cross_val_score to test recall at different depths all 3 models would have minimal difference in the max depth they were using.

In conclusion the largest factors affecting student dropout are age, attendance, stress index, study hours per day, travel time, and family income. However, I don't think that my models are fully trustworthy for predicting student dropouts. With an average recall score of .355 this means it's only catching a third of the cases in which a student drops out. Due to the low recall score and pretty low accuracy score I can't fully trust

that my models are accurate at predicting student dropout regardless of the changes made to the variables.



Looking at the visualization for this confusion matrix it is obvious that the model isn't very accurate. When it came to predicting whether a student is going to drop out or not and was often incorrect more times than not. With a large number of false positives and negatives this wouldn't be a model that is trusted in a real world situation and there are too many errors for it to be used.

If I had to do this project again I would use a dataset that wasn't a synthetic dataset as it resulted in a lot of null values that needed to be filled and a lot of interpretation with the data that otherwise would be less of a problem with a collected database. However, this can lead to a lot of response bias or a breach of privacy to ask students who dropped out personal information about themselves. I still think that an

accurate model can be good for student health as schools can use these variables to try and help students improve stress and attendance rate.

Resources:

Dataset:

https://www.kaggle.com/datasets/meharshanali/student-dropout-prediction-dataset/data

College Dropout Rates:

https://educationdata.org/college-dropout-rates